

FYKE SIMON V. TONEL

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Education

STI College Caloocan

2022 – Present

Bachelor of Science in Computer Science

Caloocan City, Metro Manila

- Current Cumulative GWA: 1.27
- Best Thesis, Bachelor of Science in Computer Science (Campus-wide – ICTE Symposium), 2024-25
- Best Methods of Research Project, Bachelor of Science in Computer Science (Campus-wide – ICTE Symposium), 2024–25
- Entrant, Codefest (Local – Tagisan ng Talino), 2024–25
- President’s Honor (STI Academic Center – Academic Honors), 2022 – Present
- Alumni Scholar (Awarded for High Honors in Senior High; 50% scholarship in college)

STI College Caloocan

2020 – 2022

Senior High School (Information & Communications Technology)

Caloocan City, Metro Manila

- Graduated with High Honors
- Completion of Computer Programming – Java and .NET Technology (IT in Mobile App & Web Development)

Lakan Dula High School

2016 – 2020

Junior High School

Manila City, Metro Manila

- Graduated with Honors

Experience

Back-End Developer (Intern)

January 2026 – February 2026

Inspire Holdings Inc.

- Designed and implemented backend systems using NestJS and AWS, improving system scalability and reliability.
- Integrated UnionBank payment APIs for secure financial transactions.
- Implemented caching and rate limiting using Redis, reducing server load.
- Developed unit tests and contributed to SDLC and QA processes.

Software Developer (Intern)

February 2026 – March 2026

Socia I.T. Solutions

Projects

LootBX Mobile: Real-Time Streaming & Interactive Rewards Platform

February 2026

React Native, Expo, TypeScript, JavaScript, MongoDB

Full-Stack Developer

- Developed API integrations supporting **real-time data flow** for a mobile streaming and rewards platform.
- Implemented state management logic during migration from **web-based architecture to unified mobile application**.
- Contributed to full-stack development of a **low-latency streaming system**, improving responsiveness and scalability.
- Applied TypeScript and modern backend patterns to improve **code reliability and maintainability**.

Papa Burger Ecosystem: Multi-Platform Restaurant Operations System

February 2026 – March 2026

React TSX, PHP, PostgreSQL, Next.js, Flutter, Dart

Full-Stack Developer

- Built front-end architecture for a **multi-module restaurant system** integrating POS, driver, and admin platforms.
- Developed core modules including **POS system, driver portal, and franchising interface**.
- Designed reusable UI components and state management for **scalable and maintainable frontend systems**.
- Integrated backend APIs to support **order processing, transactions, and operational workflows**.

Projects

Flick Player: Music Player with Custom UAC 2.0

January 2026

Flutter, Dart, Rust, Kotlin, DSP

Lead Developer – Maintainer

- Built a Rust-based audio engine implementing custom UAC 2.0, enabling **bit-perfect playback** and bypassing Android AudioFlinger for external DAC/AMP support.
- Developed a dual-engine architecture (Rust + just_audio) with automatic fallback, improving **device compatibility across USB and non-USB audio paths**.
- Designed a real-time DSP pipeline (EQ, dynamics, spatial effects, gapless, crossfade) with **low-latency audio processing**.
- Implemented Flutter–Rust FFI bridge for **thread-safe, low-overhead communication** between UI and native engine.
- Optimized music library processing (scanning, metadata extraction, indexing), improving **performance on large libraries**.
- **Open-sourced the application, reaching 600+ downloads and early user adoption.**

br41ndmg: High-Fidelity Audio Resampling Library

January 2026

Rust (DSP, Audio Processing)

Lead Developer

- Implemented polyphase sinc-based resampling for **high-accuracy sample rate conversion**.
- Developed configurable FIR filters (Hann, Hamming, Blackman, Kaiser) for **precise frequency response control**.
- Built both offline and real-time streaming pipelines for **flexible audio processing use cases**.
- Validated DSP performance using impulse, sine, and sweep testing with **Criterion benchmarks**.
- Achieved **>100 dB stopband attenuation and <0.1 dB passband ripple**.

PASADA: Ride-Hailing & Fleet Management Ecosystem

December 2024 – November 2025

Flutter, TypeScript (Bun/Deno), React, Supabase, Google Cloud Platform

Lead Full-Stack Developer

- Architected a multi-platform system connecting **4 applications** (Passenger, Driver, Admin, Web) into a unified transport ecosystem.
- Developed an analytics pipeline generating **7-day predictive forecasts** for booking demand and route traffic.
- Integrated Gemini API to transform **real-time and predictive data into actionable summaries** for users and fleet managers.
- Built and maintained Flutter mobile apps with **real-time traffic integration (Google Maps)** for route optimization.
- Developed a Next.js web platform with a **live ride-tracking system** using MapLibre GL and OpenStreetMap.

Mochi: Family-Shared AI Companion with Persistent Memory System

April 2026

Flutter, Node.js, SQLite, llama.cpp, Gemini API, Termux, Kotlin

Lead Developer

- Designed a multi-user AI system with **persistent memory, mood tracking, and relationship state modeling**.
- Built a hybrid inference pipeline combining **local LLM (llama.cpp)** and **cloud fallback (Gemini API)**.
- Developed backend services for **conversation history, memory injection, and state management**.
- Implemented real-time synchronization across devices using **WebSockets**.
- Deployed on a low-cost infrastructure (Termux + Cloudflare Tunnel) for **continuous availability**.

Locker: Advanced Media Hiding & Security Application

November 2025

Flutter, Dart, Kotlin

Lead Developer – Maintainer

- Developed an Android media vault with **biometric authentication, PIN protection, and AES-256 encryption**.
- Implemented security features including **decoy vault and forced-access protection mechanisms**.
- Built a full media pipeline (import, compression, secure storage, recovery) supporting **multiple file types**.
- Designed scalable file management (search, tagging, deduplication) for **large datasets**.
- Optimized performance (memory usage, compression, rendering) for **smooth UX under heavy load**.
- Released as an open-source application with **active real-world usage**.

Skills

Core Languages

Dart, Rust, Kotlin, TypeScript, Java, SQL

Systems & Backend Engineering

Node.js (NestJS, ExpressJS), Bun, Deno

PostgreSQL, SQLite, Supabase, Redis, QuestDB

WebSockets, API design, RBAC, rate limiting

Mobile & Frontend Development

Flutter, React, Next.js, Remix

Tailwind CSS, MapLibre GL JS

Low-level & Performance

Rust (systems programming, DSP, audio processing)

FFI (Flutter ↔ Rust), real-time pipelines, memory optimization

Cloud & Infrastructure

Docker, Vercel, Fly.io, Cloudflare, GCP

Event-driven systems (SQS/SNS), caching, deployment pipelines

AI & Applied ML

llama.cpp, Gemini API

Tools & Environments

Linux (Arch Linux and CachyOS), Neovim, Git, GitHub, GitLab, Postman

Android Studio, Neovim, Zed

Figma

Certifications

Learn Dart: First Step to Flutter

July 2024

Educative.io

Java Fundamentals

June 2023

Oracle